

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 2nd Term Regular Examination, 2019
URP 3231
(Transportation Planning)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Mention different types of walking practices in brief. What are the components of walking in urban transportation system? (15)
- (b) What is meant by walkability? Illustrate the factors of walkability with necessary examples. (10)
- (c) A walkway of 8 feet with an obstruction of 2.5 feet operates at a level of service A/B. (10)
 - (i) Find the capacity of that walkway.
 - (ii) If there are 1500 walkers per hour eastward and 1700 walkers per hour westward. Find the level of service at which it operates.
2. (a) Mention the advantages of bus transport over other forms of transport alternatives. Illustrate the role of agencies involved in bus transportation in short. (15)
- (b) In light of the principles of bus route planning what measures you will take as a transport planner to avoid overlapping of bus route pairs. (10)
- (c) A transit line employing non-articulated vehicles is expected to carry 10,000 passengers during the 2-hour morning peak period. Given a round trip of 30 min and an average vehicle occupancy of 75 passengers, Calculate the hourly flow q and the number of vehicles F required to provide this flow. (10)
3. (a) Mention the differences between Time Mean Speed and Space Mean Speed with necessary examples. (10)
- (b) Mention different types of volume counts with their specific purposes and applications. (10)
- (c) The following table gives the particulars collected for a section of road 0.7 km long during the course of a moving observer study: (15)

Direction of the Run	Journey Time (min)	Stopped Time (min)	Vehicles met with in the opposing direction			Vehicles in the same direction	
			Car	Bus	Truck	Over-taking vehicles	Over-taken vehicles
N-S	1.01	0.04	13	0	5	1	0
S-N	0.92	0.10	11	0	0	2	1
N-S	0.77	0.08	19	2	11	1	1
N-S	0.03	0.14	14	2	4	1	0
S-N	0.84	0.08	2	0	11	0	1
S-N	1.06	0.13	19	1	7	2	1

Calculate the flow in PCU/hr in both directions (N-North & S- South) of traffic assuming an equivalency factor of 3 for bus and 2 for trucks. Calculate the Journey Speed and Running Speed.

4. (a) Exemplify the following terms: (06)
 - (i) Parking Load, (ii) Parking Index, (iii) Parking Turn-over
- (b) It has been found that on an average 1 in 100 drivers in a bus company are involved in an accident every year. If there are 500 drivers in the company, what is the probability that there are exactly 4 drivers who are involved in an accident during a year? (05)

- (c) "The characteristics of a road have a great influence on the causation of accidents"- Narrate this statement in brief. (12)
- (d) Illustrate various Travel Demand Management (TDM) strategies with necessary examples. (12)

Section – B

5. (a) Explain the concept of transportation system. Discuss the components of a transportation system with examples. (15)
- (b) What are the sequential demand forecasting models? Explain with necessary examples. (12)
- (c) Define TAZ. Mention the selection criteria of TAZ. (08)
6. (a) Describe different ways for estimating the number of trip production with necessary examples. (15)
- (b) Explain the concept of trips from origin-destination and production-attraction point of view. (08)
- (c) Make a comparative analysis among different growth factor models of trip distribution with examples. (12)
7. (a) Four Zones of Khulna City have the following productions and attractions: (20)

Zone	Production	Attraction
A	1000	3000
B	2000	3000
C	3000	2000
D	4000	2000

The travel matrix is:

Zone	Travel Time (min)			
	A	B	C	D
A	2	5	7	10
B	5	3	8	12
C	7	8	2	11
D	10	12	11	03

Travel Time (min)	F_{ij}	Travel Time (min)	F_{ij}
2	3.0	8	1.2
3	2.5	10	1.0
5	2.3	11	0.95
7	1.5	12	0.90

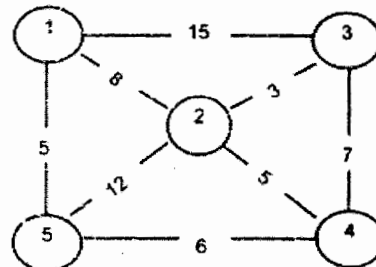
Apply the gravity model to distribute the trips ($K_{ij}=1.23$)

- (b) Let the number of trips from i to j is 9551. Three modes are available from i to j to make the trips which has the characteristics given by the following table. Compute the trips carried by the three models and the fare required to travel by each mode. (15)

	Vehicle travel time between i and j	Walking time to and from stops	Waiting time at stops	Fare charged between i and j	Parking cost
Coefficient	0.03	0.04	0.06	0.1	0.1
Car	25	-	-	15	3
Bus	35	8	4	8	-
Train	17	12	3	5	-

8. (a) What is modal split? Describe the factors influencing the modal split with examples. (15)
- (b) A trip table (veh/hr) needs to be loaded on the network shown in figure. Find the new link travel time using capacity restrained assignment. Assume capacity of each link to be 750veh/hr. The trip matrix and street network is given below (Travel time assigned on each link is in minutes) : (20)

From \ To	1	2	3	4	5
1	0	50	60	70	30
2	40	0	30	60	80
3	90	40	0	20	50
4	80	70	90	0	30
5	30	40	50	60	0



Khulna University of Engineering & Technology
Department of Urban and Regional Planning
 BURP 3rd Year 2nd Term Regular Examination, 2019

URP 3223
(Regional Development Planning)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Explain the concept of Region. Give a short overview about the different types of regions with examples. (10)
 (b) "Regional Planning is the process of formulating and clarifying social objectives in the ordering of activities in supra-urban space." Explain. (15)
 (c) What is meant by Regional delineation? Write down the role of Regional Delineation in Regional Planning. (10)
2. (a) Define Regional Planning. Discuss the Need for Regional Planning in Bangladesh. (10)
 (b) "Within a functional region there is usually a central hub or point of distribution for whatever function or activity the region is concerned with."- Justify the statement with proper examples and diagrams. (15)
 (c) Write Short note on : (10)
 (i) Gravitational analysis
 (ii) Methodologies for delineation of functional regions
3. (a) Define Regional Disparity .Describe the factors affecting Regional Disparity in the context of Bangladesh. (10)
 (b) What is Tourism Regions and Planning? Briefly discuss the necessity of Tourism Planning in Bangladesh. (15)
 (c) Write down the strategies for the Pourashavas and City Corporations under Sixth Five Year Plan 2011-2015. (10)
4. (a) Write down the necessity of revamping the property tax system for the Pourashavas and City Corporations. (05)
 (b) Write short notes on: (10)
 (i) Targets for reduction of regional disparity
 (ii) Shortcomings of shift share analysis
 (c) Briefly describe the procedure of delineating formal regions with necessary figures. (10)
 (d) Discuss the objectives of Land Use Planning. (10)

Section – B

5. (a) "Regional database plays the most vital role in regional delineation"- Explain. Prepare a list of data necessary for delineating a region efficiently. (12)
 (b) Narrate the significance of followings: (06)
 (i) Inter-regional planning
 (ii) Intra-regional planning
 (c) Elaborate the basic structure of "Input-Output" analysis with an example (assume required data). Mention the applicability of input-output analysis in regional planning. (07)
 (d) Transform the following simple transaction matrix into input-coefficient matrix, then calculate direct and indirect input requirements for the final demand 3000 for manufacturing and 1000 for commercial sector (after 3rd round calculation). (10)

Output to Input from	Intermediate Purchase	
	Manufacturing	Commercial
Manufacturing	0.50	0.20
Commercial	0.33	0.30
Household	0.17	0.50
Total	1.00	1.00

6. (a) Evaluate the applicability of central place theory considering following countries: (15)
i) Bangladesh ii) Australia
- (b) Make a comparison between the following two aspects of growth pole theory with example. (10)
i) Spread effect ii) Backwash effect
- (c) Discuss the contribution of agglomeration economies on regional growth. Illustrate the agglomeration economies given below: (10)
i) Economies internal to firm
ii) Economies external to firm but internal to industry

7. (a) Suppose, a number of jute processing industries are located at the periphery of Khulna city. Due to some difficulties it has become unavoidable to shift the location from Khulna to Mongla. Identify some indicators having relevance with the situation and briefly narrate the process of developing a decision matrix. (15)
- (b) "Gravity model of human interaction is used to determine market boundary between two centers"- How? A and B are two regions, 50 km apart from each other, having population of 200,000 and 50,000 respectively. Identify the market boundary of the regions. (12)
- (c) Specify and depict the relationship between indicators used in "probabilistic approach to the gravity model" for analyzing regional shopping behaviors. Use necessary equation. (08)

8. (a) Illustrate the contribution of regional economic models in regional growth analysis with necessary example. (05)
- (b) Critically analyze the following theories on regional growth: (21)
i) Macro-economic model
ii) Inter regional resource allocation model
iii) Export base theory
- (c) Regional growth can be either turned into "convergence" or "divergence"- discuss such scenario in regional planning with provision for intervention. (09)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 2nd Term Regular Examination, 2019
URP 3291

(Operations Research and Systems Analysis)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Necessary chart and graphs will be supplied.

Section – A

1. (a) Define Operation Research (OR). Write down the basic components of mathematical models mentioning practical example. (07)
- (b) An iron ore from 4 mines will be blended. The analysis of study has shown that, in order to obtain suitable tensile properties, minimum requirements must be met for 3 basic elements A, B, and C. Each of the 4 mines contains different amounts of the 3 elements. Formulate to find the least cost blend for one ton of iron ore. (15)

Basic Elements	Minimum Requirement	Composition from each mine lb/ton			
		1	2	3	4
A	5	10	3	8	2
B	10	90	150	75	175
C	30	45	25	20	37
Cost/ ton from each mine L.E/ ton of ore		800	400	600	500

- (c) Solve the following Linear Programming problem by graphical method – (13)

$$\max z = 6x_1 + 8x_2$$

Subject to, $5x_1 + 10x_2 \leq 60$; $4x_1 + 4x_2 \leq 40$; $x_1, x_2 \geq 0$

2. (a) Solve the following L.P.P. by the simplex method: (20)

$$\text{Minimize } z = 3x_1 + 4x_2$$

Subject to, $2x_1 + x_2 \geq 8$; $x_1 + 2x_2 \geq 8$; $x_1, x_2 \geq 0$

- (b) Briefly explain the following terms: (15)
- (i) Degeneracy (ii) Alternative Optima (iii) Infeasible Solution

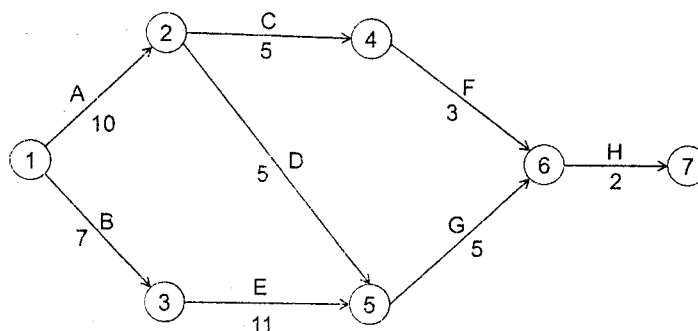
3. (a) Define network analysis. Differentiate between PERT and CPM. (08)

- (b) The following table contains information related to the major activities of a research. Use the information to do the following: (15)

- (i) Draw a precedence diagram using AOA.
- (ii) Determine the expected length of the project.

Activity	Precedes	Expected Time (Days)	Activity	Precedes	Expected Time (Days)
a	c,b	5	i	m	10
c	d	8	m	End	8
d	i	2	g	h	1
b	i	7	h	k	2
e	f	3	k	End	17
f	m	6	f	m	6

- (c) What is computing algorithm? Using computing algorithm determine the critical path for the following network of a project. (12)



4. (a) Define game theory. Write down the basic elements of a game. (07)
 (b) Solve the following game graphically - (18)

		Player B			
		B ₁	B ₂	B ₃	B ₄
Player A	A ₁	2	2	3	-2
	A ₂	4	3	2	6

- (c) Find out the saddle point of the game- (10)

		A			
		1	2	3	4
A	1	-5	-3	1	20
	2	5	5	4	6
	3	-4	-2	0	-6

Section – B

5. (a) What are meant by decision under risk and decision under uncertainty? Explain with suitable example. (08)
 (b) Consider the following payoff matrix. (15)

	S ₁	S ₂	S ₃	S ₄	S ₅
a ₁	7	9	-3	10	15
a ₂	0	2	-9	5	8
a ₃	14	27	19	-3	22
a ₄	-13	30	39	27	7

No probabilities are known for the occurrence of future states. Determine the solutions by the following criteria:

- (i) Laplace (ii) Maximin (iii) Savage (iv) Hurwicz (Assume $\alpha=0.60$)
- (c) Farmer McCoy can plant either corn or soybeans. The probabilities that the next harvest prices of those commodities will grow up, state the same, or go down are 0.35, 0.40, and 0.25 respectively. If the prices grow up, the corn crop will net \$35,000 and the soybeans will net \$15,000. If the prices remain unchanged, McCoy will (barely) break even. But if the prices go down, the corn and soybeans crops will sustain losses of \$25,000 and \$10,000, respectively. Which crop should McCoy plant? (12)
6. (a) What is queue? Write some important applications of queuing theory. (05)
 (b) Briefly explain the basic components of a queue system. (10)
 (c) Customers arrive at a sale counter manned by a single person according to a Poisson process with a mean rate of 20 per hour. The time require to serve a customer has an exponential distribution with mean of 100 seconds. Find- (12)
 (i) the probability that the counter is busy
 (ii) the average waiting time of a customer
 (iii) the average number of customers in the queue
 (d) Write down a brief description of the various types of queue system. (08)
7. (a) Briefly explain the following terms: (12)
 (i) Absorbing State; (ii) Transient State; (iii) Recurrent State; (iv) Periodic State
 (b) What is meant by limiting probability? Explain. (05)
 (c) An urn always contains 3 balls. Ball colors are red and blue. We can say that the system is in state i , $i = 0, 1, 2, 3$, if the urn contains i red balls. At each stage a ball is randomly chosen and then replaced by a new ball, which with probability 0.8 is the same color, and with probability 0.2 is opposite color, as the ball it replaces. Write down the detailed calculation to find its transition probability matrix. (18)
8. (a) What is meant by simulation? What are the advantages and limitations of simulation? (10)
 (b) In a certain city, if today is sunny, tomorrow will be sunny 70% of the time. If today is cloudy, tomorrow will be cloudy 60% of the time. Suppose, today is Friday and the weather is cloudy. What is the probability that the weather will be sunny on Tuesday? (10)
 (c) Patients arrive at a 1-doctor clinic according to a Poisson distribution at the rate of 15 patients per hour. The waiting room does not accommodate more than 10 patients. Examination time per patient is exponential with mean rate 12 per hour. (15)
 (i) Find the effective arrival rate at the clinic.
 (ii) What is the probability that an arriving patient will get service immediately?
 (iii) What is the expected total time a patient spends in the clinic?

Khulna University of Engineering & Technology
 Department of Urban and Regional Planning
 BURP 3rd Year 2nd Term Regular Examination, 2019
CE 3207
 (Elements of Civil Engineering Structures)

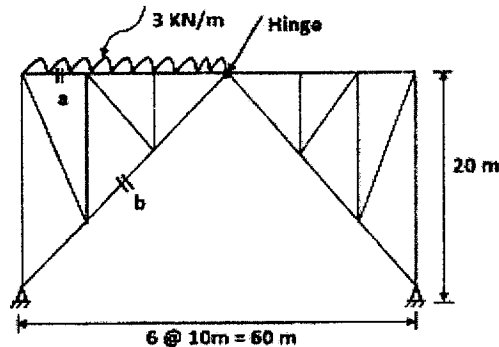
Full Marks: 210

Time: 3 hrs

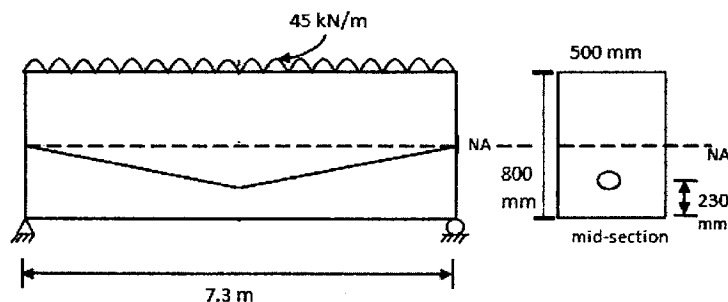
- N.B.**
- Figures in the right margin indicate full marks.
 - Assume reasonable value for any missing data.
 - Answer any three questions from each section in separate script.

Section – A

- Define determinate, indeterminate and stable structure with suitable example (10)
 - What is meant by truss structure. Write down the name and draw five common types of truss. (07)
 - Compute the bar forces for "a" and "b" members in the following truss as shown in figure below and indicate whether it is in compression or tension. (18)

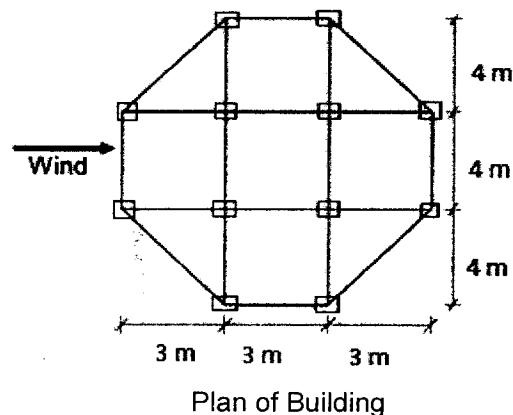


- What is prestress concrete ? Explain in details the three concept of prestress concrete. (12)
 - Enumerate the advantages of prestress concrete over RCC. (08)
 - For the following member, calculate the top fiber and bottom fiber stress due to prestress force of 1620 kN by the 1st and 2nd concept at the mid section. (15)

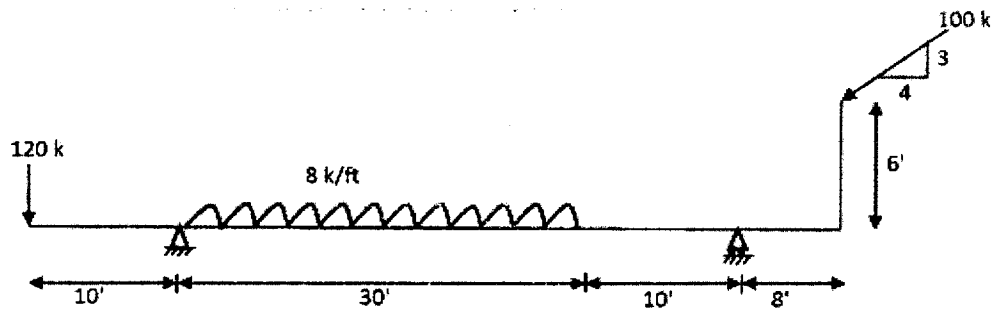


- Calculate the earthquake load at each floor level for an Eight storied RCC school building located at Khulna. The seismic dead load for the first four storey is 3000 kN/Floor, for 4th and 5th storey 2600 kN/Floor and 2200 kN/Floor for all other stories. Consider the height of each floor as 5m. (20)
 - Calculate the design wind load at each floor level for a 5 - storied building from the following data: (15)
 - Height of each floor = 3m
 - Basic wind speed = 170 Km/hr
 - Building located at Khulna
 - Exposure category 'C'
 - Pressure co-efficient = 1.50

Height (m)	C_G	C_z
3	1.321	0.801
6	1.249	0.866
9	1.258	0.972
12	1.233	1.055
15	1.215	1.125



4. (a) Write down the characteristics of an ideal truss. (05)
 (b) Draw the shear force and bending moment diagram for the structure shown in the figure below. (30)



Section – B

- 5 (a) Define Reinforced Cement Concrete (RCC) . What are the advantages of RCC as a structural material ? (08)
 (b) What is modular ratio? Write down the significance of modular ratio. (07)
 (c) A simply supported rectangular beam with a span of 18 ft supports a uniform live load of 900 lb/ft, dead load of 400 lb/ft and a concentrated load of 500 lb at the middle of the span. If $f'_c = 4000$ psi and $f_y = 60,000$ psi, determine the required cross-section and steel area. Also show the reinforcement details. (20)
- 6 (a) What is a slab ? According to ACI code, what are the minimum thickness of slab with different support conditions? (05)
 (b) A RCC slab is built integrally with its supports and consists of three equal spans each with a clear span of 18 ft. The working live load is 60 psf and 4000 psi concrete is specified for use with steel yield stress equal to 50,000 psi. Design the one way slab. (30)
- 7 (a) Define column. How can you fix the bar spacing in tied column ? (10)
 (b) Design a circular spiral reinforced column for a concentric allowable working load of 320 kips with $f'_c = 3,000$ psi and $f_y = 60,000$ psi (25)
- 8 (a) Define foundation. What are the different types of footing? (05)
 (b) A column 12" x 18" supports a dead load of 130 kips and a live load of 150 kips. The allowable soil pressure is 4 ksf. Design a single column footing and show the reinforcement details if $f'_c = 3500$ psi and $f_y = 60,000$ psi. (30)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 2nd Term Regular Examination, 2019
CE 3261
(Water Resources Planning)

Full Marks: 210

Time: 3.0 hrs.

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Briefly describe the fluvial system of an idealized river-course from its source to outfall. (10)
 - (b) Describe the mechanics of formation of a meandering river from a straight one. What are the variables that govern the meandering process? (09)
 - (c) Differentiate between: (16)
 - (i) Tidal and deltaic river
 - (ii) Rocky and Boulder river stage
 - (iii) Aggrading and Degrading river
 - (iv) Sinuous and Non-sinuous braided river
 2. (a) Explain the sediment transport mechanism in a river. (07)
 - (b) What is river navigation? Explain its importance in the modern system of transport. (08)
 - (c) What is meant by the term "river training"? What are the different objectives served by it? (08)
 - (d) What are the methods available to train a river in Bangladesh? Explain any three methods of the name. (12)
 3. (a) What is flood? Why flood occurs so frequently in Bangladesh? - Discuss with geographical location. (10)
 - (b) What is meant by flood control? What are the different measures of flood control and which one is most suitable for Bangladesh and why? (12)
 - (c) A reservoir has been planned with a useful life of 100 years. The storage capacity of the reservoir is 1000 Mm³. The average annual yield from the basin is 10,000 Mm³. The average annual sediment flow to be 2 Mm³. In how many years, the reservoir capacity will be depleted 50% of its present capacity due to silting? (13)
- | | | | | | | |
|-------------|-----|------|------|------|------|------|
| (C/I) ratio | 0.1 | 0.09 | 0.08 | 0.07 | 0.06 | 0.05 |
| Te (%) | 86 | 84 | 82.5 | 81 | 78 | 76 |
4. (a) Define irrigation. What are the advantages and ill-effects of irrigation? (09)
 - (b) Write short note on: (i) Furrow irrigation method (ii) Sprinkler irrigation method (iii) Drip irrigation method. (12)
 - (c) What is meant by delta of a crop? Show the relation between duty and delta. (08)
 - (d) If wheat requires about 8.9 cm of water after every 30 days and the base period for wheat is 180 days, find out the delta for wheat. (06)

Section – B

5. (a) Define hydrologic cycle. What are the importance of hydrologic cycle? Draw hydrologic cycle with flow diagram. (12)
- (b) Define infiltration. Explain which factors affect the infiltration process. (08)
- (c) What is evaporation? Describe the factors affecting the role of evaporation of an area. (10)
- (d) Write down the difference between evaporation and transpiration. (05)

6. (a) What is precipitation? What are the conditions to form precipitation? (05)
- (b) Describe the energy budget method for determination of the lake evaporation (10) and show that, $E_L = \frac{H_n - H_g - H_s - H_L}{\rho L (1 + \beta)}$, where the symbols have their usual meaning.
- (c) A reservoir with a surface area of 250 hectares had the following average (10) value of parameters during a week: Water temperature= 20°C, relative humidity= 40%, wind velocity at 1.0 m above ground= 16 km/h. Estimate the average daily evaporation from the lake and value of water evaporated from the lake during that week.
- (d) What is 'Φ- index'? A storm with 10 cm precipitation produce a direct runoff of (10) 5.8 cm. Give the time distribution of the storm as below, estimate the 'Φ- index' of the storm.

Time from Start (h)	1	2	3	4	5	6	7	8
Incremental rainfall in each hour (cm)	0.4	0.9	1.5	2.3	1.8	1.6	1.0	0.5

7. (a) What factors should be considered in selecting a site for a stream gauging (07) station.
- (b) Define unit hydrograph. Draw a typical hydrograph and show its different (08) components.
- (c) The following are the data obtained in a steam-gauging operation. A current (20) meter with a calibration equation of $V = (0.32N + 0.032)$ m/s; where, N= revolutions per second was used to measure the velocity at 0.6m depth. Using the mid-section method, calculate the discharge in the stream.

Distance from the bank (m)	0	2	4	6	9	12
Depth (m)	0	0.50	1.10	1.95	2.25	1.85
Number of revolutions	0	80	83	131	139	121
Time (s)	0	180	120	120	120	120
Distance from left bank (m)	15	18	20	22	23	24
Depth (m)	1.75	1.65	1.50	1.25	0.75	0
Number of revolutions	114	109	92	85	70	0
Time (s)	120	120	120	120	150	0

8. (a) Write down the uses of unit hydrograph. Describe briefly the basic (08) assumptions that constitute the foundation of unit hydrograph theory.
- (b) Following are the ordinates of a storm hydrograph of a river draining a (15) catchment area of 423 km² due to a 6h isolated storm. Derive the ordinate of a 6-h unit hydrograph for the catchment.

Time from start of storm (h)	0	6	12	18	24	30	36	42	48
Discharge (m ³ /s)	10	30	87.5	115.5	102.5	85	71	59	47.5
Time from start of storm (h)	54	60	66	72	78	84	90	96	-
Discharge (m ³ /s)	39	31.5	26	21.5	17.5	15	12.5	12	-

- (c) Give the ordinate of a 4-h unit hydrograph as below; derive the ordinates of a (12) 12-h unit hydrograph for the same catchment by 'S curve method'.

Time (h)	0	4	8	12	16	20	24	28	32	36	40	44
Ordinate of 4-h UH (m ³ /s)	0	20	80	130	150	130	90	52	27	15	5	0